

求级数 $\sum_{n=1}^{\infty} \frac{(1+\frac{x}{n})^n}{n^x}$ 的收敛域

$$u_n(x) = \frac{(1+\frac{x}{n})^n}{n^x} \sim \frac{e^x}{n^x} = \frac{1}{n^x e^{-x}} \quad (n \rightarrow \infty)$$

$$\lim_{n \rightarrow \infty} \frac{u_n(x)}{\frac{1}{n^x}} = e^x > 0$$

根据比较判别法 当 $x > 1$ 时 $\sum_{n=1}^{\infty} \frac{1}{n^x}$ 收敛. 故原级数收敛

当 $x \leq 1$ 时 $\sum_{n=1}^{\infty} \frac{1}{n^x}$ 发散. 故原级数发散

$$\int_0^1 \frac{\log(1+x)}{x} dx \quad \text{收敛性}$$

$$\int_0^1 \frac{1}{x} \cdot \log(1+x) = \int_0^1 (1 - \frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{4} + \dots) dx$$

$$= 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$$

$$= (1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots) - 2(\frac{1}{2^2} + \frac{1}{4^2} + \frac{1}{6^2} + \dots)$$

$$= \frac{\pi^2}{6} - \frac{1}{2}(1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots)$$

$$= \frac{\pi^2}{12}$$

$$\int_0^{2\pi} \log(1 - 2r\cos\theta + r^2) d\theta = 0 \quad |r| < 1$$

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}, \quad z = r \cdot e^{i\theta}$$

$$\sum_{n=0}^{\infty} r^n \cos n\theta = \frac{1 - r\cos\theta}{1 - 2r\cos\theta + r^2}$$

$$\sum_{n=1}^{\infty} r^n \cos n\theta = \frac{r\cos\theta - r^2}{1 - 2r\cos\theta + r^2}$$

$$\int_0^t \sum_{n=1}^{\infty} r^{n-1} \cos n\theta dr = \int_0^t \frac{\cos\theta - r}{1 - 2r\cos\theta + r^2} dr = -\frac{1}{2} \int_0^t \frac{1}{1 - 2r\cos\theta + r^2} d(1 - 2r\cos\theta + r^2)$$

$$\sum_{n=1}^{\infty} \frac{r^n}{n} \cos n\theta = -\frac{1}{2} \log(1 - 2r\cos\theta + r^2)$$

$$\sum_{n=1}^{\infty} \frac{r^n}{n} \int_0^{2\pi} \cos n\theta = 0$$